

Ve 216: Introduction to Signals and Systems

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Based on Lecture Notes by Prof. Jeffrey A. Fessler and Prof. Yong Long

Outline

- 1 8. Communications
 - Introduction
 - Sinusoidal amplitude modulation (8.1)
 - Demodulation (8.2)
 - Frequency-division multiplexing (8.3)

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 - Asynchronous Demodulation
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Introduction

This chapter applied FT principles and tools to the analysis of **communications systems** (AM radios, FM radios, wireless phones, video, etc.)

All of it is based on the **modulation property** of the FT.

Overview

- Amplitude Modulation (AM radio, digital comm (modems))
- Synchronous demodulation
- Asynchronous demodulation
- Heterodyning tuner

We will only cover 8.1, 8.2, 8.3

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Modulation Property

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Definition

Modulation is the process used to shift the frequency of an information signal so that the resulting signal is in the desired frequency band.

Importance of modulation

- 1 *Ear can hear from about 20Hz to 20kHz. For such audio signals, electrical transmission over copper wire works fine (and is used in all audio systems). So there is no need to modulate.*
- 2 *Human voice is dominated by frequency components of less than 1kHz. If we were attempt to transmit a human voice signal by the propagation of electromagnetic (radio) waves, we could encounter the several problems. Two of the more obvious problems are as follows.*

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Antenna length requirement (1)

1. Antenna length requirement.

For efficient radiation, antenna lengths should be longer than $\lambda/10$, where λ is the wavelength of the signal to be radiated, given by

$$\lambda = \frac{c}{f_c}$$

where c is the speed of the light and f_c is the frequency of the signal.

Example

For a 30Hz frequency component, the wavelength is

$$\lambda = \frac{c}{f_c} = \frac{3 \cdot 10^8 \text{ m/s}}{30 \text{ s}^{-1}} = 10,000 \text{ km},$$

so $\lambda/10 = 1000 \text{ km}$. Obviously impractical.

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Interference from other signals

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- If all radio stations transmitted baseband 20Hz-20kHz signals (over a cable for example), the receivers would receive all of those signals **superimposed**, and would have **no way of separating them**.
- The solution to these problems is to **modulate signals** to **different parts of the frequency spectrum**.

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FCC regulation

In the U.S., the **frequency band allocation** is controlled by the Federal Communications Commission (FCC).

- **AM radio**: 540-1600kHz
- **FM radio**: 88-108MHz

Note: lots of room for many non-overlapping channels each $\pm 20\text{kHz}$ wide.

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DSB/SC-AM

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- We have already seen that the **modulation property** of the FT provides a mechanism for this.
- There are **many variations** on how to do this (see Ve 353 Introduction to Communication Systems).
- The simplest method is called **double sideband, suppressed carrier, amplitude modulation** or **DSB/SC-AM**.

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Sinusoidal amplitude modulation (1)

Let $x(t)$ be the **modulating signal** (the audio signal) and let $c(t)$ be the **carrier signal** (a high-frequency cosinusoid). Specifically

$$c(t) = \cos(\omega_c t + \theta_c)$$

where ω_c is called the **carrier frequency**.

The transmitted signal is

$$y(t) = x(t)c(t) = x(t)\cos(\omega_c t + \theta_c)$$
$$\xleftrightarrow{\mathcal{F}} Y(\omega) = \frac{1}{2}[e^{j\theta_c}X(\omega - \omega_c) + e^{-j\theta_c}X(\omega + \omega_c)]$$

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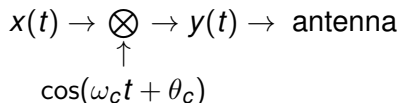
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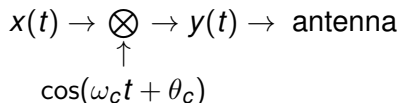
Block diagram of modulation system:



(Picture) in frequency domain (MIT, Lecture 13.9) (textbook, Figure 8.4, p. 586) This multiplication is also sometimes called **mixing**. Note that graphically it appears that **all of the audio signal information is retained**, it is just **moved to a different part of the spectrum**.

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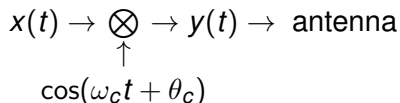
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Since it is now a **high frequency** signal, it can be transmitted by a **practical antenna**, and **different radio stations can use different ω_c 's**.

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*40kHz would be a **bare minimum**, but that would require an **ideal filter** to separate the different channels. So somewhat **wider separation** desirable.*

Sinusoidal amplitude modulation (4)

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The carriers of AM radio stations are spaced by 10kHz! How?

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Smaller bandwidth, so lower audio quality.

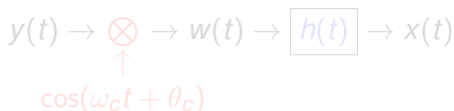
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Synchronous Demodulation

- The signal must be **restored to baseband** for our **ears** to hear it.
- Conceptually, this can be done by **another multiplication** by a $\cos(\omega_c t + \theta_c)$ signal, followed by **lowpass filtering**.

Block diagram of demodulation system

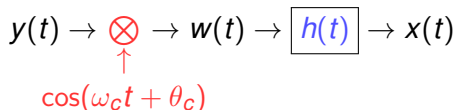


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Frequency response and lowpass filtering

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$$W(\omega) = \frac{1}{2} [e^{j\theta_c} Y(\omega - \omega_c) + e^{-j\theta_c} Y(\omega + \omega_c)]$$

$$= \frac{1}{4} e^{2j\theta_c} X(\omega - 2\omega_c) + \frac{1}{2} X(\omega) + \frac{1}{4} e^{-2j\theta_c} X(\omega + 2\omega_c)$$

$$Y(\omega) = \frac{1}{2} [e^{j\theta_c} X(\omega - \omega_c) + e^{-j\theta_c} X(\omega + \omega_c)]$$

Now pass through a **lowpass filter** to extract the baseband signal. (**Picture**)(MIT, Lecture 13.10 and 13.11)

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*Yes! At your receiver you do not have the modulation signal $\cos(\omega_c t + \theta_c)$ available. Specifically, the **phase** θ_c is not available.*

Loss of synchronization in phase (1)

Effect of loss of synchronization in phase. (Picture)(MIT, Lecture 13.16) (textbook, Figure 8.9)

- 1 If the phase difference is $\pi/2$, the output will be *zero*.
- 2 If the amplitude is small and there exists noise in the system, the *signal to noise ratio is small*.
- 3 If the phase relation between θ_c and ϕ_c is not maintained over time, the *amplitude factor* $\cos(\theta_c - \phi_c)$ *varies*.

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Synchronous demodulation requires a sophisticated phase-tracking receiver.

- *An expensive receiver is reasonable in some applications like satellite communications where transmit power is limited, so an expensive base station is acceptable.*
- *But in applications like commercial AM radio, one would like the receivers to be inexpensive, even if that means a more complicated and power-inefficient transmitter.*

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A simple system and associated waveform for an **asynchronous demodulation** system. **(Picture)**(MIT, Lecture 13.17)

- The **envelop** of $y(t)$ (a smooth curve connecting the peaks) is a reasonable approximation of $x(t)$.
- $x(t)$ could be approximately recovered through the use of the **envelop detector** that tracks these peaks to extract the envelop.

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DSB/WC-AM

Definition

Transmitting some of the modulation signal too (uses power) is called **double sideband, with carrier, amplitude modulation** or **DSB/WC-AM**.

Now the modulated signal (transmitted) is:

$$y(t) = (A + x(t)) \cos(\omega_c t)$$

(**Picture**) of block diagram (MIT, Lecture 13.18) (textbook, Figure 8.12) Spectrum of the DSB/WC signal:

$$Y(\omega) = A\pi[\delta(\omega - \omega_c) + \delta(\omega + \omega_c)] + \frac{1}{2}[X(\omega - \omega_c) + X(\omega + \omega_c)]$$

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As A is increased

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- Since the envelope of the DSB/WC signal describes the original audio signal, an envelop detector (**(Picture)**, MIT, Lecture 13.17) will recover the envelope.

$$y(t) = (A + x(t)) \cos(\omega_c t)$$

$$m(t) = A + \hat{x}(t)$$

- Just eliminate the DC component and you recover the original signal.

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Frequency-division multiplexing (1)

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- Commercial AM: **10KHz spacing** between carriers.
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- **(Picture)** of block diagram multiple transmitters (MIT, Lecture 13.12)
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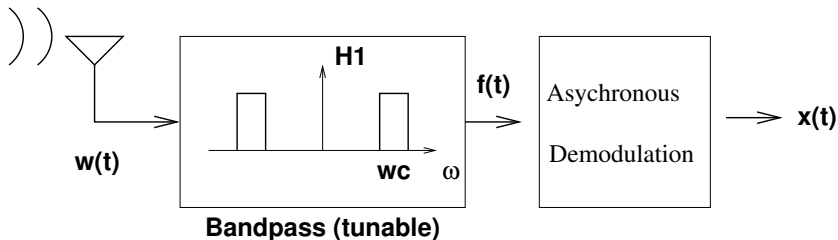
Tunable bandpass filter (1)

Question

How can we “tune in” to our favorite station?

Design 1: a tunable bandpass filter.

Tuning (Design #1)



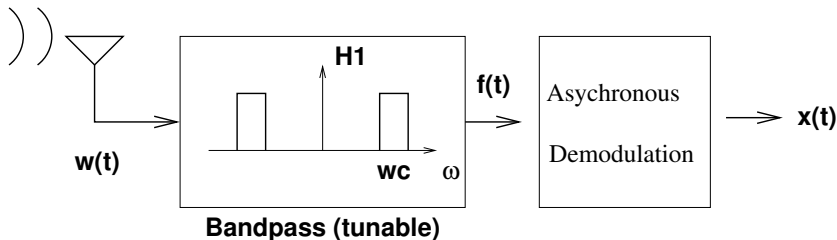
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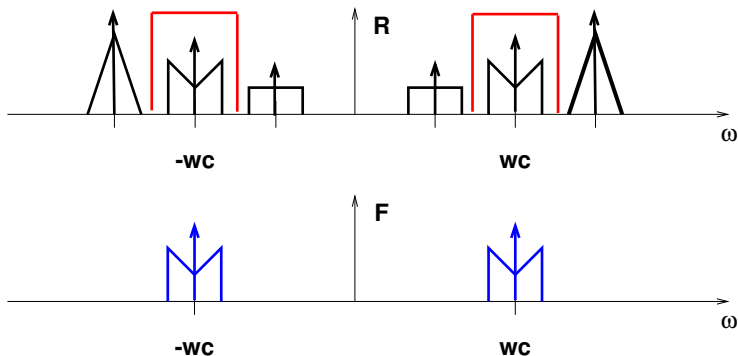
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Tunable bandpass filter (2)

(Frequency Division Multiplexing)



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- ① narrowband (± 5 kHz)
- ② high center frequency (540 to 1600 kHz)
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- ⑤ sharp cutoff (to avoid interference from neighboring channels)

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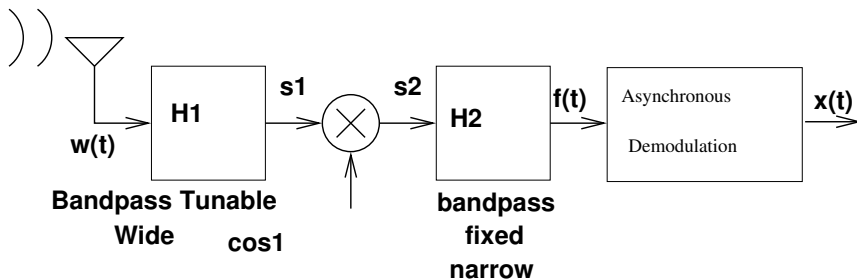
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- 5 sharp cutoff (to avoid interference from neighboring channels)

*These concurrent requirements are difficult to achieve in practice, so **commercial AM radios are not built this way.***

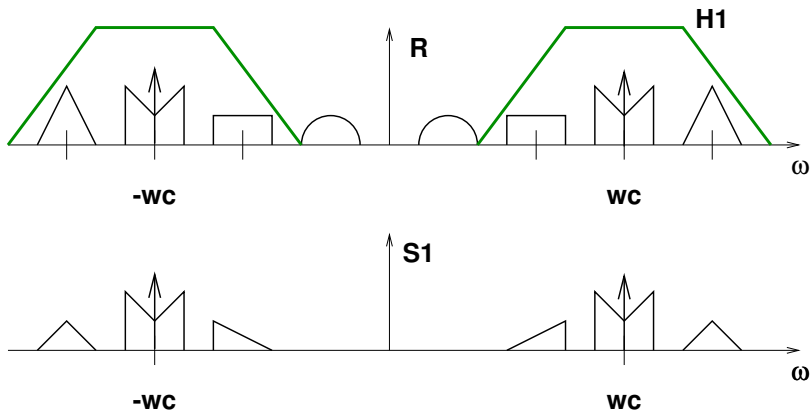
Superheterodyning receiver

Design 2: **superheterodyning receiver** (avoids a tunable narrow bandpass filter)

Superheterodyning Tuner



Bandpass tunable wide



IF filter and mixing frequency

Definition

The fixed frequency bandpass filter called the **IF filter** for “intermediate frequency.”

In commercial AM this is $\omega_{\text{IF}}/2\pi = 455\text{kHz}$.

Modulate the received RF signal by $\omega_0 = \omega_c + \omega_{\text{IF}}$ to move the spectrum of the desired signal to be centered at ω_{IF} .

$$s_1(t) \rightarrow \begin{array}{c} \otimes \\ \uparrow \\ \cos(\omega_c + \omega_{\text{IF}}) t \end{array} \rightarrow s_2(t)$$

When you “tune in” you are actually controlling the **mixing frequency** ω_0 .

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Mixing frequency: example

Example

If we want to tune in to the station at

$$\omega_c/2\pi = 1060\text{kHz}$$

then use

$$\omega_0/2\pi = (\omega_c + \omega_{\text{IF}})/2\pi = 1060 + 455 = 1515\text{kHz}.$$

Before mixing, centered at $\pm 1060\text{kHz}$.

After mixing by 1515kHz , centers at

$$\pm\omega_c \pm \omega_0 \implies \pm\omega_{\text{IF}} \text{ and } \pm(2\omega_c + \omega_{\text{IF}})$$

$$\pm 1060 \pm 1515\text{kHz} \implies \pm 455\text{kHz} \text{ and } \pm 2575\text{kHz}$$

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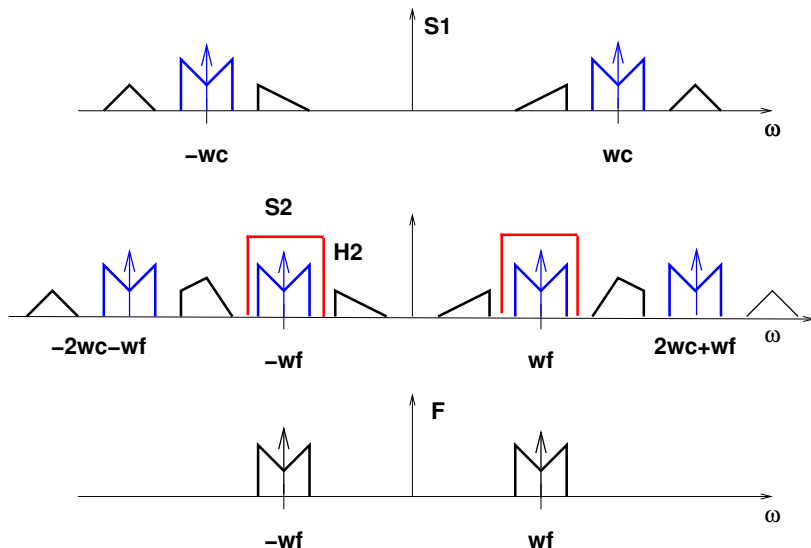
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Modulation and IF filter



Recovering the original signal

- After mixing with $\cos((\omega_c + \omega_{IF}) t)$, the desired station is centered in the **passband** of the **IF** filter.
- After the IF filter, **asynchronous demodulation** can be used to recover the original audio signal.

Question

One point we have omitted though. In the above example, what happens to the part of the received signal spectrum centered at

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Interfering spectrum components

After mixing by 1515kHz, this part gets moved to

$$\pm 1970 \pm 1515\text{kHz} \implies \pm 455 \text{ and } \pm 3485\text{kHz}.$$

So that signal also would leak through the IF filter and interfere with the desired signal.

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How to solve this problem?

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The desired signal, centered at 1060kHz, and the interfering signal, centered at 1970kHz, are separated by

$$1970\text{kHz} - 1060\text{kHz} = 910\text{kHz} = 455\text{kHz} + 455\text{kHz} = 2\omega_{\text{IF}}/2\pi.$$

*We just need a **tunable bandpass filter** that is **centered at ω_c** , but with a passband that is about **900kHz wide**. It is fairly easy to build such a tunable bandpass filter.*

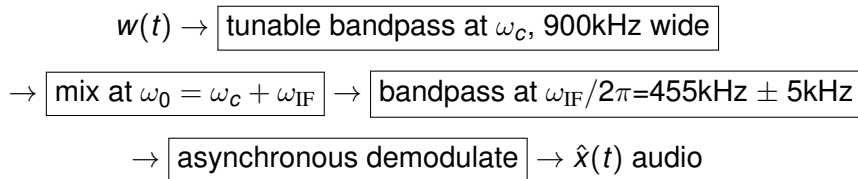
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Consider the effects of loss of synchronization in phase between the carrier signal and the mixing signal.

A better approach? (2)

To simplify the analysis, let's consider one station only. The modulated (transmitted) signal is

$$y(t) = (A + x(t)) \cos(\omega_c t + \theta_c)$$

After mixing at the receiver

$$\begin{aligned} s_2(t) &= y(t) \cos((\omega_c + \omega_{IF}) t + \phi_c) \\ &= \frac{1}{2}(A + x(t)) \cos((2\omega_c + \omega_{IF}) t + \theta_c + \phi_c) \\ &\quad + \frac{1}{2}(A + x(t)) \cos(\omega_{IF} t + \theta_c - \phi_c) \end{aligned}$$

If $\omega_{IF} = 0$, then after passing a lowpass filter,

$$\hat{x}_c(t) = \frac{1}{2}(A + x(t)) \cos(\theta_c - \phi_c)$$

See the effects of the additional term $\cos(\theta_c - \phi_c)$ on Chap. 8, p.49.

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Summary

- double sideband, suppressed carrier, amplitude modulation (DSB/SC-AM)
- double sideband, with carrier, amplitude modulation (DSB/WC-AM)
- synchronous demodulation
- asynchronous demodulation
- Frequency-division multiplexing (superheterodyning receiver)
- Systems-level analysis of communication system